

VISHNU MP

Kerala, India (Open to relocate to Germany)

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Summary

Control Systems Engineer and Industrial Automation Architect with 9+ years of experience designing distributed control systems, PLC architectures, and scalable industrial software solutions across aerospace and process industries. Expertise in IEC 61131-3 and IEC 61499 standards, SCADA and HMI engineering, and multi-vendor automation ecosystems including Siemens, Rockwell, and Schneider Electric. Proven track record delivering safety-critical automation systems for ISRO and global industrial clients. Strong focus on modular automation, Industry 4.0 integration, and OPC UA-based interoperability. Currently advancing capabilities in Artificial Intelligence and Machine Learning to enable intelligent control systems, predictive maintenance, and data-driven industrial operations.

Core Skills

- **Control Systems Engineering:** Distributed Control Systems (DCS), PLC Programming, Real-Time Control Systems, Safety PLC Configuration, Control Systems Architecture
- **Standards & Programming:** IEC 61131-3 (ST, FBD), IEC 61499, OPC Unified Architecture (OPC UA)
- **SCADA & Visualization:** SCADA Systems, High Performance HMI (ISA-101), Alarm Management, SIMATIC WinCC, Aveva System Platform
- **Industrial Communication Protocols:** Profinet, Profibus, Ethernet/IP, ControlNet, Modbus, HART, RS-485, RS-232
- **Automation Platforms:** EcoStruxure Automation Expert (EAE), TIA Portal, Studio 5000, Codesys, Pilz PAS4000
- **Industry 4.0 & IIoT:** Industrial Internet of Things (IIoT), Smart Manufacturing, Digital Transformation, System Integration
- **AI & Intelligent Systems:** Machine Learning, NLP, Computer Vision, Predictive Analytics, AI-assisted Engineering Tools
- **Leadership & Delivery:** Agile Methodology, Global Stakeholder Management, Technical Mentoring, Cross-functional Collaboration

Experience

Schneider Electric

May 2024 – Present

Staff Senior Design Engineer

Bengaluru, India

- Define architecture standards and development strategy for IEC 61499-based PLC application libraries within EcoStruxure Automation Expert, enabling scalable and vendor-agnostic automation solutions.
- Architect reusable function block libraries supporting modular, event-driven control across process, hybrid, and discrete manufacturing environments.
- Develop AI-enabled internal engineering tools for version control automation and template standardization, improving development efficiency and reducing repetitive engineering effort.
- Collaborate with global OEMs, system integrators, and R&D teams to deliver Industry 4.0 solutions integrating SCADA, HMI, and IIoT ecosystems using OPC UA and modern automation frameworks.

Schneider Electric

May 2022 – April 2024

Senior Design Engineer

Bengaluru, India

- Develop standardized IEC 61499 automation libraries enabling reusable and modular control system design across global industrial deployments.
- Design SCADA and HMI components with real-time visualization, alarm management, and equipment state monitoring for distributed automation systems.
- Perform gap analysis between modern automation platforms and legacy PLC systems, designing migration strategies to ensure seamless system transition.
- Execute unit and integration testing across distributed control environments ensuring compliance with industrial automation standards and system requirements.

Axcend Automation and Software Solutions**July 2021 – April 2022***Senior Application Engineer**Bengaluru, India*

- Develop IEC 61499 based control and supervision application libraries using EcoStruxure Automation Expert for industrial automation solutions.
- Support functional validation and automated testing of automation library components ensuring reliable deployment across control systems.
- Deliver automation software development using Agile methodology supporting iterative releases and engineering collaboration.

Proficient Automation and Controls Pvt. Ltd.**May 2018 – June 2021***Senior Project Engineer**Kerala, India*

- Lead end-to-end automation design and commissioning for aerospace systems including ISRO Gaganyaan engine test facilities, delivering safety-critical PLC and SCADA systems.
- Architect multi-vendor control systems integrating Siemens, Rockwell, and Schneider PLCs with industrial communication networks and SCADA platforms.
- Implement high-availability control architectures with redundant PLC and SCADA systems ensuring reliability in mission-critical environments.
- Conduct FAT and SAT procedures validating system performance, safety compliance, and operational readiness for industrial and aerospace applications.

Kerala State Electronics Development Corporation (KELTRON Projects)**May 2017 – April 2018***Automation Engineer – Industrial Training and Technical Support**Kerala, India*

- Deliver industrial automation training programs for 50+ engineers covering PLC programming, SCADA systems, and industrial communication protocols.
- Design automation solutions translating customer requirements into control system architectures and technical specifications.
- Develop technical documentation and troubleshooting frameworks improving system usability and reducing recurring support issues.

Education

Birla Institute of Technology, Pilani**Expected 2027***Master of Technology in Artificial Intelligence & Machine Learning**Rajasthan, India***Birla Institute of Technology and Science, Pilani****2020 – 2022***Master of Technology in Digital Manufacturing**Rajasthan, India***KELTRON & Government Engineering College****2016 – 2017***Post Graduate Diploma in Industrial Automation and Instrumentation Engineering**Kerala, India***MEA Engineering College, Calicut University****2012 – 2016***Bachelor of Technology in Electrical and Electronics Engineering**Kerala, India***Certifications**

- Certification in Codesys Platform and PLC Programming – PROSOFT Systems (GP Systems), Yekaterinburg, Russia, 2019
- Advanced Automation with Siemens S7-1500 (TIA Portal V15.1) – SITRAIN Siemens, Bangalore, India, 2018
- International Conference on Computing, Communication and Energy Systems – MEA Engineering College, Kerala, India, 2016
- Multiple Internal Certifications – Schneider Electric, 2026

Languages

- English – Full Professional Proficiency (C1)
- German – Intermediate Proficiency (B1)